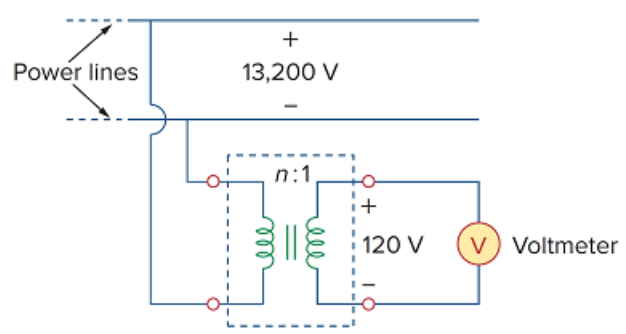


Problem

Refer to Fig. 13.61. Calculate the turns ratio required to step down the 14.4-kV line voltage to a safe level of 120 V.

Figure 13.61: A transformer providing isolation between the power lines and the voltmeter.



Step-by-step solution

Step 1 of 3

Refer to Figure 13.61 from the textbook.

For the step-down transformer, write the expression for the turns ratio, n .

$$n = \frac{V_1}{V_2} \dots\dots (1)$$

Here,

V_1 is the primary side line voltage,

V_2 is the secondary side line voltage.

Step 2 of 3

Since the line voltage to be step-down is the primary side voltage.

Thus,

$$\begin{aligned} V_1 &= 14.4 \text{ kV} \\ &= 14000 \text{ V} \end{aligned}$$

Since the line voltage is step-down to a safe value of 120 V.

Thus,

$$V_2 = 120 \text{ V}$$

Step 3 of 3

Substitute 14.4 kV for V_1 and 120 V for V_2 in equation (1).

$$\begin{aligned} n &= \frac{14000}{120} \\ &= 120 \end{aligned}$$

Therefore, the required turns ratio, n is 120.